

WCH QingKe RISC-V cores summary

Core	ISA	Branch pred.	Interrupts	HPE levels	Int. nesting	VTF channels	Pipeline	Vector table mode	I-cache	XW	FPU	PMP areas	Priv. modes
V2A	RV32EC	static	256	2	2	2	2	address / instruction		✓			M
V2C	RV32EC_Zmmul	static	256	2	2	2	2	address / instruction		✓			M
V3A	RV32IMAC	static	256	2	2	4	3	Instruction					M, U
V3B	RV32IMCB	static	256	2	2	4	3	address / instruction		✓			M, U
V3C	RV32IMCB	static	256	2	2	4	3	address / instruction		✓		4	M, U
V3F	RV32IMACFB	static	256	2	2	4	3	address / instruction		✓	✓		M, U
V4A	RV32IMAC	BHT / BTB / RAS	256	2	2	4	3	address / instruction				4	M, U
V4B	RV32IMAC	BHT / BTB / RAS	256	2	2	4	3	address / instruction		✓			M, U
V4C	RV32IMAC	BHT / BTB / RAS	256	2	2	4	3	address / instruction		✓		4	M, U
V4F	RV32IMACF	BHT / BTB / RAS	256	3	8	4	3	address / instruction		✓	✓	4	M, U
V4J	RV32IMAC	BHT / BTB / RAS	256	2	2	4	3	address / instruction	64K	✓		4	M, U
V5F	RV32IMACFB	static	256	8	8	4	7-9	address / instruction		✓	✓	4	M, U

BHT Branch History Table

BTB Branch Target Buffer

RAS Return Address Stack

HPE Hardware Prologue / Epilogue

VTF Vector Table Free interrupt response mechanism

XW Compressed instructions for self-extending byte and half-word operations (proprietary extension)